

Session Objectives

1. Differentiate the pathogenesis and development of thoracic aortic aneurysms (TAA), aortic abdominal aneurysms (AAA) and aortic dissections.
2. Compare and contrast TAA vs. AAA in their distinct clinical features, risk factors, diagnostic measures, management, and complications
3. Discuss the clinical presentation, diagnosis, classifications and management of aortic dissections.

Lecture Outline

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Anatomy Review

- Aorta and Blood Vessel

What is an Aneurysm?

- Terminology + Classification
- Pathogenesis
- Risk Factors

Thoracic Aortic Aneurysm (TAA)

Abdominal Aortic Aneurysm (AAA)

- Risk factors
- Clinical Presentation & Diagnosis
- Complications
- Management

Aortic Dissection

- Pathogenesis & Classifications
- Clinical Presentations & Diagnosis
- Management

Conclusions

- Summary of lecture content



TAA – Risk Factors

Degenerative Changes

- **Hypertension**
- Atherosclerosis
- Hypercholesterolemia
- Cigarette smoking
- Infection/Inflammation (syphilis, Staph Aureus)
- Aortitis
- Chest Trauma (ex. car accident)

HyperTension is the most important risk factor for TAA formation!

Congenital Risk Factors (TAA > AAA)

- Marfan Syndrome -> A.D. defect in **fibrillin**
- Ehlers-Danlos Type IV -> A.D. defect in **type-III collagen**
- Loeys-Dietz -> A.D. defect in **TGF-β receptor**
- Bicuspid Aortic Valve



Traumatic Aortic Injury (TAI)

Traumatic rupture of the aorta
due to **acute deceleration**
injury.

“Pseudoaneurysm”

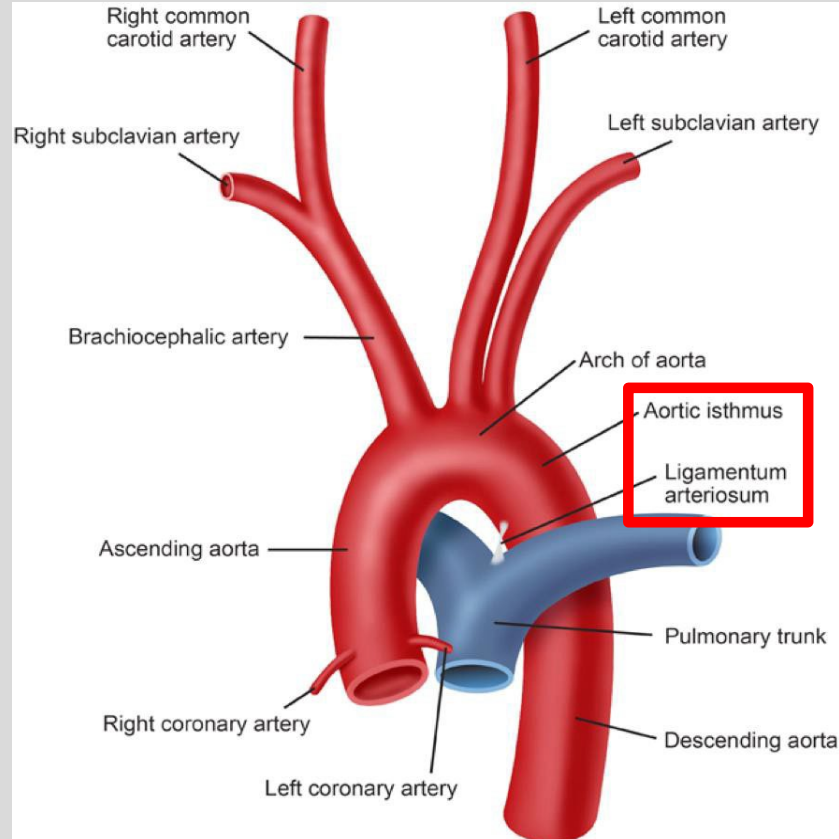
-> High-speed (30+ mph) MVCs or
falls from 10+ feet

**Second leading cause
of traumatic death!**

Most common site is **aortic
isthmus** (distal to left subclavian
artery).

- Why is this??

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UpToDate: Clinical
Features & Diagnosis
of Blunt Thoracic
Aortic Injury

Marfan Syndrome (MFS)

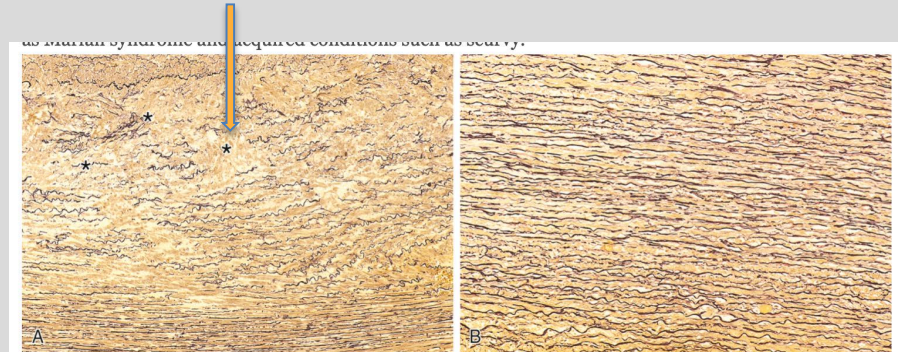
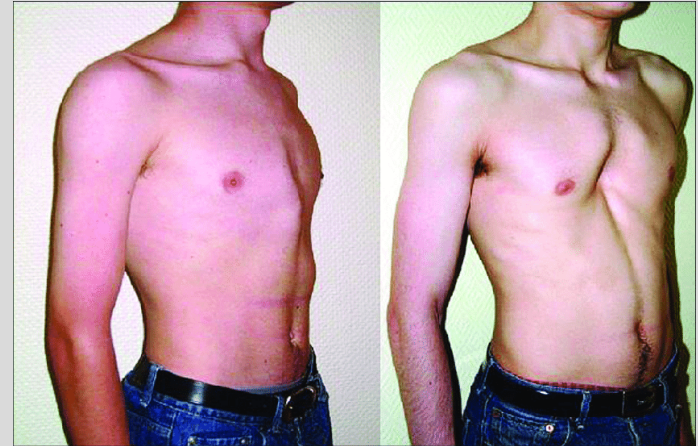
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Pathogenesis of Aneurysms

- **Autosomal dominant** defect in the fibrillin gene (chromosome 15)
 - **FBN1 gene mutation**
- **Cystic medial necrosis of aorta**
- Skeleton, eyes, and cardiovascular system = most affected
- **Most common cause of death = aortic root aneurysm rupture or dissection**

Clinical Characteristics

- Wingspan > height
- Long, slender extremities
- **Pectus excavatum or carinatum**
- Upwardly-dislocated lenses
- Mitral valve prolapse
- Scoliosis



Robbins & Cotran Pathologic Basis of Disease, 10th edition

UpToDate: Genetics, clinical features, and diagnosis of Marfan syndrome and related disorders

Ehlers-Danlos Syndrome Type IV (Vascular EDS)

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Pathogenesis of Aneurysms

- Autosomal dominant defect in type-III collagen synthesis
- Type III collagen forms a supportive **mesh** within vessels
- **80% of patients experience a major vascular event or rupture of an internal organ by 40 years old**
 - **Most common cause of death = arterial rupture**
- Life expectancy = 48 years
- At least 1 in 100,000 people (4% of all EDS cases)

Clinical Characteristics

- Thin, translucent skin that bruises easily
- Hypermobility
- Acrogeria (premature aging)
- Large eyes; thin face, lips, and nose
- Poor wound healing



Loeys-Dietz Syndrome (LDS)

Pathogenesis of Aneurysms

- Autosomal dominant defect in **TGF- β receptor**
- Present with aneurysms much earlier than Marfan & EDS-IV

Clinical Characteristics

- *Often mistaken for Marfan syndrome*
- Features differentiating from Marfan syndrome:
 - Bifid uvula (LDS until proven otherwise)
 - Hypertelorism (90%) = wide-set eyes
 - Cervical spine deformity

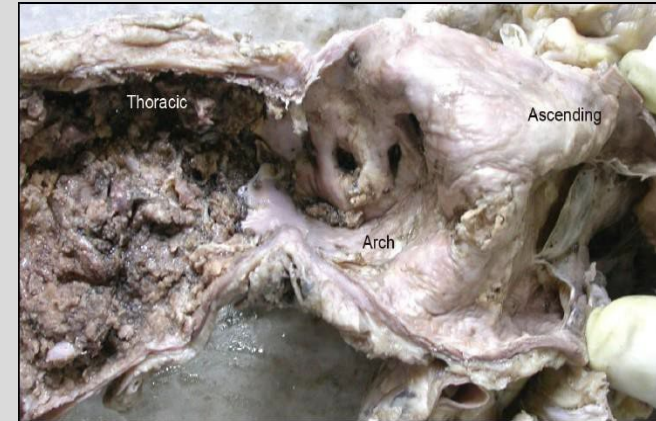


Syphilitic Aortitis

Primary infection of the aortic wall is rare

- Typically 2/2 to another process (endocarditis, IVDA, sepsis)
- Syphilis (*T. pallidum* **endarteritis** of the **vasa vasorum**)
 - Clinical manifestations can occur ~15-30 years after initial infection
 - Results in **obliterative endarteritis** -> interruption of blood flow -> medial ischemia -> loss of elastic & muscular elements -> ANEURYSM

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Syphilitic aortitis -> 3° syphilis and can lead TAA in a patient



TAA – Clinical Presentation

TAA's are usually **silent** unless a complication occurs, or an imaging study has been performed for another reason

- Complications = **rupture** or dissection

The most common presenting symptom of TAA is pain in chest or abdomen.

TAA's can affect surrounding mediastinal structures w/ resulting symptoms:

- **Ascending Thoracic Aorta**
 - Aortic regurgitation due to dilation of aortic root & annulus (can lead to heart failure)
- **Aortic Arch**
 - Coronary artery -> myocardial ischemia or infarction
 - Superior vena cava -> JVD, upper extremity & face edema
 - Pulmonary artery -> pulmonary embolism-like symptoms
 - Airway -> dyspnea, cough, wheezing
 - Left recurrent laryngeal nerve -> hoarseness
- **Descending Thoracic Aorta**
 - Aneurysm eroding into spine -> Back pain

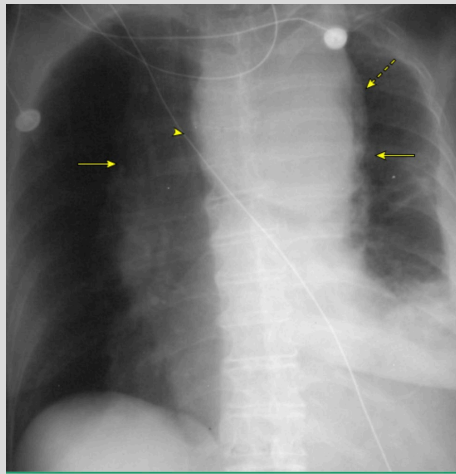


TAA – Diagnosis

Many are discovered incidentally on imaging performed for other reasons!

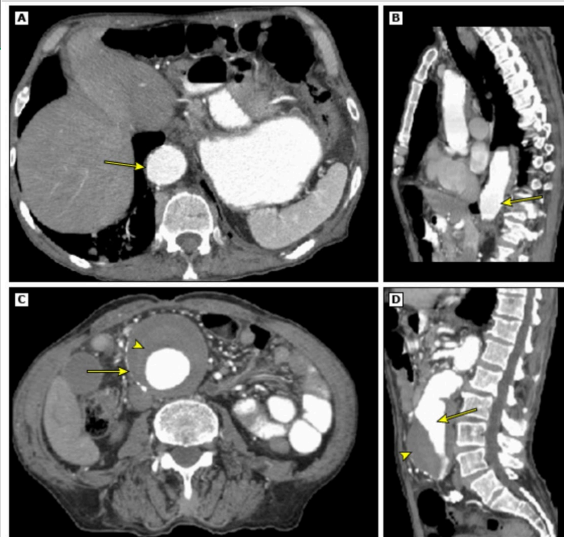
Best initial test (if stable) → CXR

- Widened mediastinum
- Enlargement of aortic knob
- Deviation of trachea



Gold standard → CT-angiogram (CTA)

- Can visualize entire aorta
- **3-D reconstruction/mapping**
- Operative planning
- Can be performed quickly in ED



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TAA – Management w/o Repair

Imaging

- CT angiography
- **Every 6 or 12 months**
- Assessing for other aneurysms
- Elective repair not recommended if TAA below certain diameter

20% of patients diagnosed with TAA have AAA

Genetic testing -> TAA can be associated with genetic diseases

If patient is asymptomatic (and diameter <5.5cm):

- Reduce cardiovascular risk
- Manage hypertension (BB > ACE or ARB)



TAA – Management w/ Repair

Repair and Surgical Intervention is indicated if:

- Asymptomatic patient with diameter > **5.5cm**
- Rapid expansion of TAA (>0.5cm/year)
- MFS, LDS, BAV or other inherited disorders
- Patients undergoing aortic valve surgery

If patient is symptomatic:

- **Surgical intervention (non-ruptured or not)**
- Open or endovascular surgical approach
 - **Open** often requires cardiac bypass
 - **Endovascular** associated with lower perioperative mortality

Marfan syndrome (MFS), familial thoracic aortic aneurysm/dissection (FTAAD), others*	
Without risk factors ^{¶Δ}	≥5.0 cm
With risk factors ^{¶Δ}	≥4.5 cm
Loeys-Dietz syndrome (LDS)	
Without risk factors ^{¶Δ}	≥4.5 cm
With risk factors ^{¶Δ}	≥4.0 cm
Bicuspid aortic valve (BAV)	
Without risk factors ^{¶Δ}	≥5.5 cm
With risk factors ^{¶Δ}	≥5.0 cm
Concomitant aortic valve procedure	≥4.5 cm



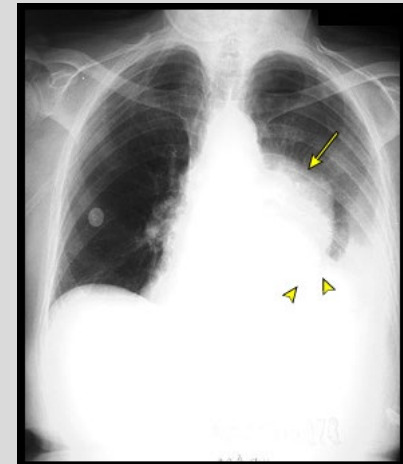
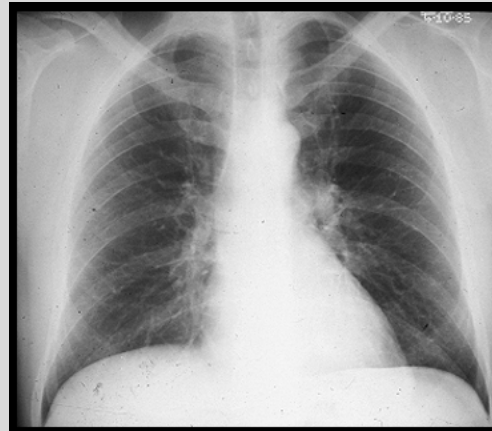
Complications – Rupture

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Patients are **hypotensive** & experience **severe pain** based on location:

- **Anterior chest** → ascending aorta
 - Rupture causes back-bleed into pericardial sack → **acute pericardial tamponade**
- **Upper back** → thoracic aorta
 - Blood pours into pleural cavity
 - Pleural effusion on CXR
 - Hemorrhagic shock
 - Respiratory failure
- **Aortoesophageal fistula** -> esophagus
 - Hematemesis

Mortality is very high in patients that experience a ruptured TAA





Question Break #1

A 60-year-old man presents to your office for his annual physical. He states that overall he feels fine, he just sometimes gets heart palpitations. He currently takes 20mg of Atorvastatin. Vital signs reveal BMI 30, BP 150/95, pulse 90bpm, temp. 96.8 F, and RR 18 bpm. Office EKG was within normal limits so you order an ultrasound, and incidentally find a thoracic aortic aneurysm (TAA) with a diameter measuring 6.5 cm.

Which of the following would be the next best step?

- A) Refer to vascular surgery for pre-operative assessment
- B) Increase patient's statin and start on aspirin
- C) Recommend lifestyle changes such as walking program
- D) Follow-up within 6 months
- E) Reassurance

AAA – Clinical Features

Like TAAs, most patients with AAA remain **asymptomatic**.

- Physical exam *may* reveal a **pulsatile abdominal mass**.
- Many AAAs are discovered incidentally

Symptomatic AAAs may present as abdominal, flank or back pain

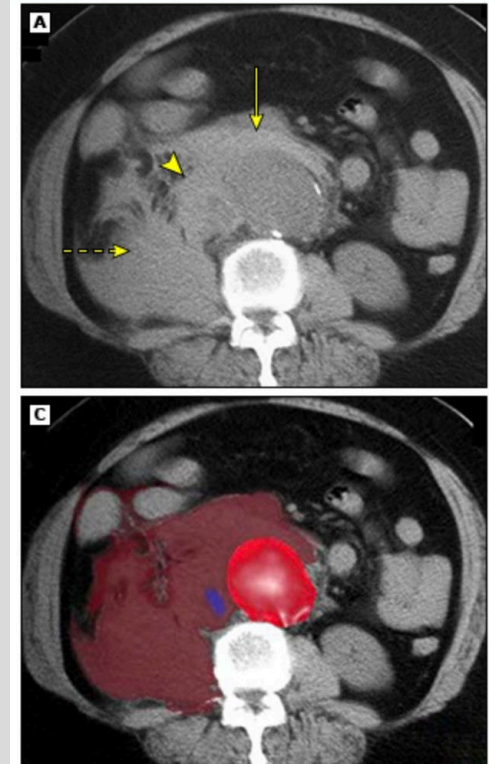
- Complications: May compress local structures, distal embolization

Ruptured AAAs present with the triad of:

- Hypotension
- Flank pain → retroperitoneal irritation due to blood spillage
- Pulsatile abdominal mass

Ruptured AAAs can be contained by retroperitoneum → prolonged survival → chance of repair if caught early

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AAA Management

- Indication for repair is determined based on **size** & presence of **symptoms**
 - >5.5cm in men; >5cm in women
 - Rapid expansion ≥ 0.5 cm over 6 months or ≥ 1 cm per year
- **Non-ruptured, asymptomatic patient**
 - Manage **hypertension** and **atherosclerosis**
 - Monitor AAA size and growth with regular ultrasounds
- **Non-ruptured, symptomatic patient**
 - Assessing risk for rupture and if patient is a candidate
- **Ruptured, symptomatic patient**
 - Abdominal CT aortography before surgery if patient is hemodynamically stable
 - Endovascular vs. open surgical repair

Aortic Dissection – Pathogenesis

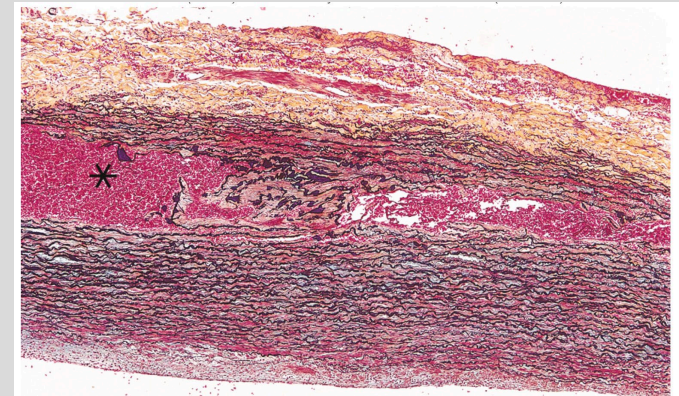
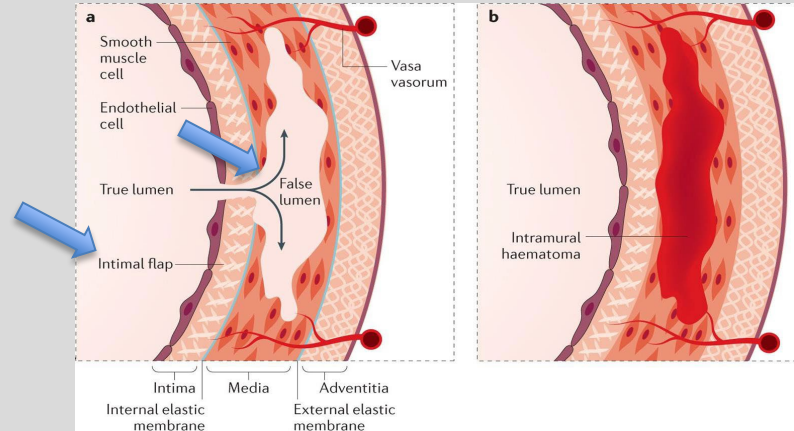
Tearing of intima and inner media from rest of aortic wall, with blood filling in the resulting space.

Why does hypertension contribute?

- Hypertension -> **degenerative changes** with loss of smooth muscle cells and altered ECM content -> **weaker** aorta walls -> tear

Atherosclerosis is NOT a part of the dissection process

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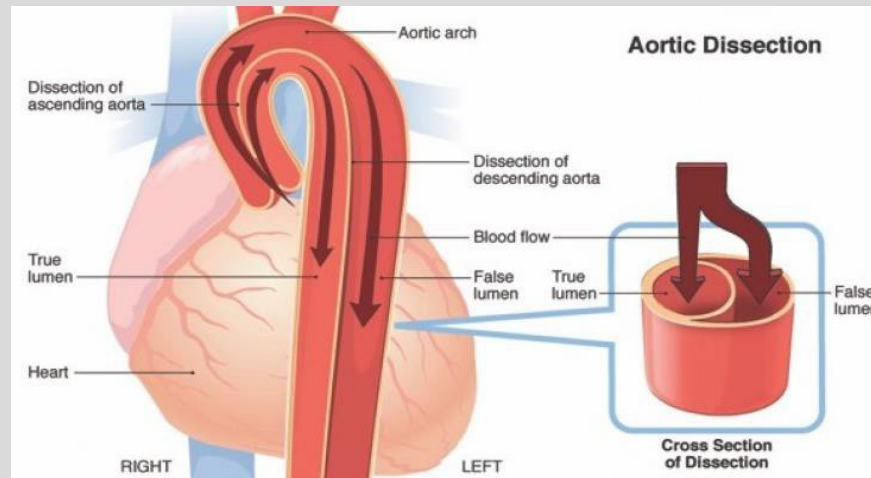


Aortic Dissection – Pathogenesis

Chronic hypertension is the major risk factor for dissection.

Tearing of intima and inner media from rest of aortic wall.

- New cavity is formed → **false lumen**, with can propagate distally **AND** proximally





Aortic Dissection – Risk Factors

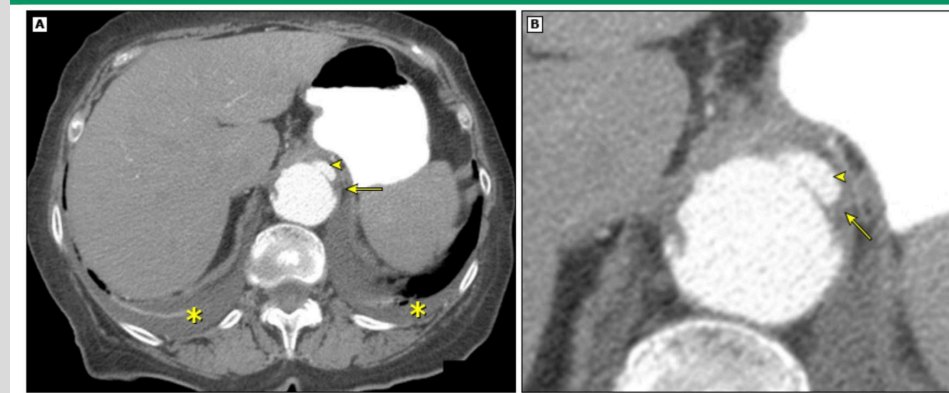
- (1) Men 40-60 years old with chronic hypertension
- (2) Younger patients with connective tissue disorders
 - *Dissection = leading cause of death in Marfan Syndrome*

Additionally:

- Dissections can be iatrogenic; ex. During cardiac catheterization, cardiopulmonary bypass

Aortic Dissection – Clinical Presentation & Diagnosis

- Sudden onset of **severe chest pain**, described as **“tearing”** with **radiation to the back**
 - Pain often severe enough to be confused as myocardial infarction
 - Dissection can propagate into coronaries -> MI
- **Discrepancy in BP** (>20 mmHg) or **pulse in extremities** -> propagation of dissection into subclavian artery



Initial **CXR** will show **widened mediastinum**.

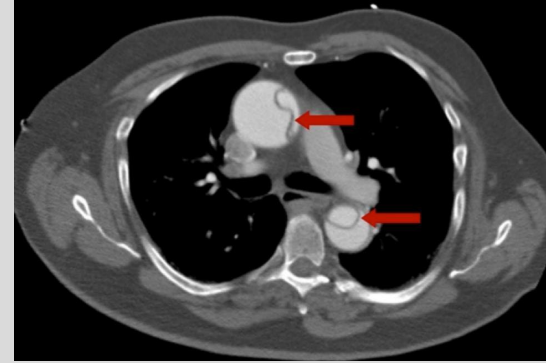
Diagnosis is confirmed with **CT-angiogram (CTA)** if possible.

Trans-esophageal echo (TEE) can be performed emergently.

Aortic Dissection – Classification & Treatment

Stanford Type A involves the
Ascending aorta

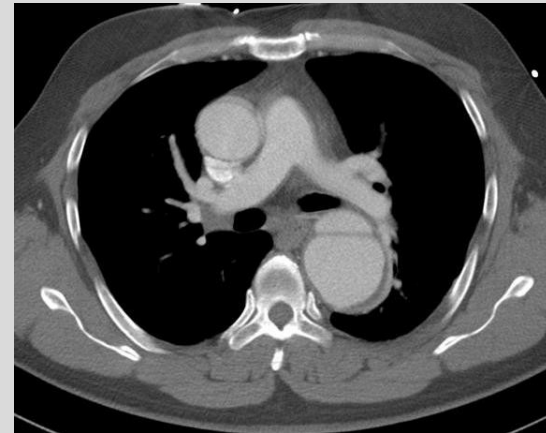
- Can involve the entire aorta
- Pericardium & coronaries involved
- Treat with **emergent surgery**



Type A Dissection

Stanford Type B does **NOT** involve the ascending aorta

- Distal to the left subclavian artery
- Treat initially with **Beta blockers**
 - Goal is to lower BP and prevent propagation
 - Vasodi**L**ators **L**ater
- Surgery is indicated with worsening vitals or complications



Type B Dissection

Final Overview

Aneurysms – General Characteristics & Pathophysiology

- Dilation of vessel wall
 - True = all three layers
 - Fusiform – circumferential/concentric; saccular - eccentric
 - False = adventitia intact
- Related to atherosclerosis -> chronic inflammation -> wall weakness
- Matrix metalloproteinases (MMPs) degrade collagen and elastin
- Cystic medial degradation is the primary pathophysiology

Thoracic Aortic Aneurysm (TAA)

- **Risk Factors:** Hypertension, genetically-mediated or congenital
- Syphilitic aortitis (3° syphilis) -> TAA 2/2 vasa vasorum endarteritis
- Usually asymptomatic, may compress mediastinal structures
- Aortic regurgitation due to dilation of aortic root + valve annulus
- Distal embolization may occlude downstream vessels
- Rupture is almost always fatal
- Diagnosis with CXR and/or CTA; typically found incidentally

Final Overview (cont.)

Abdominal Aortic Aneurysm (AAA)

- **Risk factors:** Atherosclerosis, smoking, older age, male sex, family history
- Inflammatory and mycotic AAA= macrophages & *S. aureus*, respectively
- AAAs are typically asymptomatic until rupture or discovered incidentally
- Ruptured AAA (rAAA) = hypotension, flank pain, pulsatile abdominal mass
- Ultrasound used to screen, CTA for mapping & planning of intervention

Aortic Dissection

- Chronic hypertension (in older men) is the leading cause
- Can occur in younger patients with connective tissue disease (i.e. Marfan)
- Tearing of intima + inner media = formation of intimal flap + false lumen
- Presents with chest pain that radiates to the back + pulse/BP deficit + widened mediastinum on CXR
- Stanford Type A = involves ascending → emergent surgery
- Stanford Type B = no ascending involvement → lower BP (BBs) & monitor
- Diagnose with CXR, CTA, or TEE (if patient is too unstable for CT scan)
- Can rupture → tamponade; Propagates → AR, MI, CVA, renal failure, etc.